

$$1) (r+1)^2 - r^2 = (r+1-r)(r(r+1) + r^2) \\ = (r+1)^2 + r(r+1) + r^2$$

$$(r+1)^3 - r^3 + r^2 - (r-1)^3 + \dots - 1$$

$$= (r+1)^2 + r(r+1) + r^2 + r^2 + r(r-1) + (r-1)^2 + (r-1)^2 + (r-1)(r-2) \\ + (r-2)^2 + \dots$$

$$= 2 \sum_{k=1}^{r+1} k^2 - (r+1)^2 - 1 + \sum_{k=1}^r k(k+1) = (r+1)^3 - 1$$

$$2 \sum_{k=1}^{r+1} k^2 + \sum_{k=1}^r k^2 + \sum_{k=1}^r k - (r+1)^2 - 1 = (r+1)^3 - 1$$

$$3 \sum_{k=1}^r k^2 + \frac{1}{2}r(r+1) + (r+1)^2 - 1 = (r+1)^3 - 1$$

$$3 \sum_{k=1}^r k^2 = (r+1)^3 - (r+1)^2 - \frac{1}{2}r(r+1)$$

$$6 \sum_{k=1}^r k^2 = (r+1)(2(r+1)^2 - 2(r+1) - 1)$$

$$\sum_{k=1}^r k^2 = \frac{1}{6}(r+1)(2r^2 + 4r + 2 - 3r - 2)$$

$$\sum_{k=1}^r k^2 = \frac{1}{6}r(r+1)(2r+1)$$

2)

i) Base case $n=1$:

$$\frac{1}{3}(4-1) = 1, \text{ true}$$

Assume this is true for some $n=k$, $k \in \mathbb{Z}^+$,
consider $n=k+1$:

$$\begin{aligned}\sum_{i=1}^{k+1} (2i-1)^2 &= \frac{1}{3}(4k^3 - k) + (2k+1)^2 \\&= \frac{1}{3}(4k^3 - k) + (4k^2 + 4k + 1) \\&= \frac{1}{3}(4k^3 + 12k^2 + 12k - k + 3) \\&= \frac{1}{3}(4(k+1)^3 - (k+1)), \text{ true}\end{aligned}$$

By induction, this is true for all $n \in \mathbb{Z}^+$

ii) Base case $n=1$:

$$1^3 + 5 = 6, \text{ true}$$

Assume this is true for $n=k$, $k \in \mathbb{Z}^+$, consider $n=k+1$

$$\begin{aligned}(k+1)^3 + 5(k+1) &= k^3 + 3k^2 + 3k + 1 + 5k + 5 \\ &= k^3 + 5k + 3(k^2 + k + 2)\end{aligned}$$

$$6 \mid k^3 + 5k, \text{ consider } 3(k^2 + k + 2)$$

$$2 \mid k^2 + k \text{ for even and odd } k \Leftrightarrow 6 \mid 3(k^2 + k + 2)$$

By induction, $6 \mid n^3 + 5n$ for all $n \in \mathbb{Z}^+$

iii) Base case $n=1$:

$$2^8 + 3^3 = 35, \text{ true}$$

Assume this is true for $n=k$, $k \in \mathbb{Z}^+$, consider $n=k+1$

$$2^{k+3} + 3^{2k+3} = 2 \cdot 2^{k+2} + 9 \cdot 3^{2k+1}$$

$$= 2(2^{k+2} + 3^{2k+1}) + 7 \cdot 3^{2k+1}$$

$$7 \mid 2^{k+2} + 3^{2k+1}, 7 \mid 7 \cdot 3^{2k+1} \Leftrightarrow 7 \mid 2^{k+3} + 3^{2k+3}$$

By induction, $7 \mid 2^{n+2} + 3^{2n+1}$ for all $n \in \mathbb{Z}^+$

3)

$$i) \sum_{r=1}^n (2r-1)^2 = \sum_{r=1}^{2n} r^2 - \sum_{r=1}^n (2r)^2$$

$$= \frac{1}{6} \cdot 2n(2n+1)(4n+1) - 4 \cdot \frac{1}{6} n(n+1)(2n+1)$$

$$= \frac{1}{3} n(2n+1)(4n+1 - 2n-2)$$

$$= \frac{1}{3} (2n^2 + n)(2n-1)$$

$$= \frac{1}{3} (4n^3 - n)$$

$$ii) \text{ consider } n^3 + 5n + 6 = (n+1)(n^2 - n + 6)$$

$$= (n+1)(n(n-1) + 6)$$

$$= 6(n+1) + (n-1)n(n+1)$$

$6 \mid 6(n+1)$, and $n-1, n, n+1$ consecutive integers, so $6 \mid (n-1)n(n+1)$

$$6 \mid n^3 + 5n + 6 \Leftrightarrow 6 \mid n^3 + 5n$$

$$iii) 7 \mid 4^k - (-3)^k, \text{ let } k = 2n+1$$

$$\Leftrightarrow 7 \mid 4^{2n+1} + 3^{2n+1} \Leftrightarrow 7 \mid 2^{4n+2} + 3^{2n+1}$$

$$\Leftrightarrow 7 \mid 2^{3n} \cdot 2^{n+2} + 3^{2n+1}$$

$$\Leftrightarrow 7 \mid 8^n \cdot 2^{n+2} + 3^{2n+1}$$

$$7 \mid 8^n - 1 \Leftrightarrow 7 \mid 2^{n+2} + 3^{2n+1}$$

4) induction step fails when $n=1$

no overlap between h_1 and h_2 , cannot deduce h_1 and h_2 have the same colour

5) 1, 1, 2, 3, 5, ...

$$a) F_{n+k} = F_k \cdot F_{n+1} + F_{k-1} \cdot F_n$$

Base case $k=2$, trivial

$$\text{Base case } k=3, F_{n+3} = F_{n+2} + F_{n+1}$$

$$= 2F_{n+1} + F_n, \text{ true}$$

Assume this is true for $k=r$ and $k=r+1$, $r \in \mathbb{Z}$, $r \geq 2$

consider $k=r+2$

$$F_{n+r+2} = F_{n+r+1} + F_{n+r}$$

$$= F_{r+1} \cdot F_{n+1} + F_r \cdot F_n + F_r \cdot F_{n+1} + F_{r-1} \cdot F_n$$

$$= F_{r+2} \cdot F_{n+1} + F_{r+1} \cdot F_n, \text{ true}$$

By induction, true for all $k \in \mathbb{Z}$, $k \geq 2$

$$\text{let } k=n+1, F_{2n+1} = F_{n+1}^2 + F_n^2$$

$$F_{n+2}^2 - F_n^2 = (F_{n+2} + F_n)(F_{n+2} - F_n)$$

$$= F_{n+1} \cdot F_{n+2} + F_{n+1} \cdot F_n$$

$$= F_{2n+2}$$

Base case $k=1$, trivial

Assume this is true for $k=r$, $r \in \mathbb{N}$, consider $k=r+1$

$$F_{(r+1)n} = F_{rn} \cdot F_{n+1} + F_{rn-1} \cdot F_n, \text{ which is a multiple of } F_n$$

By induction, this is true for all $k \in \mathbb{N}$

if F_n is prime, then $\gcd(F_n, F_{n-k}) = 1 \quad \forall k \leq n-1$

so using previous parts, n should be a prime

$F_6 = 2$ also a prime, so n is prime or $n=4$

b) the pair (f_n, f_{n+1}) has finite amount of possibilities

so eventually there will be a pair $(f_k, f_{k+1}) = (f_n, f_{n+1})$

$$\therefore f_{k+2} = f_{k+1} + f_k \text{ mod } 10$$

$\therefore f_n$ is periodic

b)

$$a) n! = a^2 - b^2$$

$$n! = (a+b)(a-b)$$

$n=1$, doesn't work,

$n!$ is even for $n \geq 2$

so $a+b$ or $a-b$ must be even

$$\therefore a, b \in \mathbb{Z}$$

$\therefore a+b$ and $a-b$ must both be even

$\therefore n!$ must have more than one factor of 2

$$\therefore n! = a^2 - b^2, \text{ for } n \geq 4$$

$$b) n = p_1^{a_1} p_2^{a_2} p_3^{a_3} \dots, \text{ where } p_i \text{ is prime and } a_i \in \mathbb{Z}, a_i \geq 0$$

$$\prod_{\substack{i \in \mathbb{N}, j \in \mathbb{N}, \dots \\ (i, j, \dots) \neq (a_1, a_2, \dots)}} p_i^i p_j^j \dots \mid (n-1)!$$

$$n \mid \prod_{\substack{i \in \mathbb{N}, j \in \mathbb{N}, \dots \\ (i, j, \dots) \neq (a_1, a_2, \dots)}} p_i^i p_j^j \dots \Leftrightarrow n \mid (n-1)!$$

$\therefore n \mid (n-1)!$ for non-prime n

c) we need 2025 of factor 10, more factors of 2 than 5, so count factors of 5

$$\left\lfloor \frac{n}{5} \right\rfloor + \left\lfloor \frac{n}{25} \right\rfloor + \left\lfloor \frac{n}{125} \right\rfloor + \left\lfloor \frac{n}{625} \right\rfloor + \dots$$

I wrote a python script and gets $n = 8120$

7)

a) Assume $\sqrt[3]{4}$ is rational,

$$\sqrt[3]{4} = \frac{p}{q}, \quad p, q \in \mathbb{N}, \quad p, q \text{ are coprime}$$

$$4 = \frac{p^3}{q^3}$$

$$p^3 = 4q^3, \quad p^3 \text{ is even, let } p = 2m$$

$$8m^3 = 4q^3$$

$$2m^3 = q^3, \quad q^3 \text{ is even, } q \text{ also even.}$$

p, q even contradicts with coprime

$\sqrt[3]{4}$ irrational

Assume $\log_3 4$ rational

$$\log_3 4 = \frac{p}{q}, \quad \dots$$

$$4 = 3^{\frac{p}{q}}$$

$$4^q = 3^p$$

$$\gcd(4, 3) = 1$$

impossible for $p, q \in \mathbb{Z}$

$\log_3 4$ irrational

$$b) 11n - 3m = \sqrt{11} m - 3\sqrt{11} n$$

$$\Leftrightarrow (11 + 3\sqrt{11})n = (\sqrt{11} + 3)m$$

$$\Leftrightarrow \frac{m}{n} = \sqrt{11}$$

$$3 < \sqrt{11} < 4$$

$$\Rightarrow 8n < m < 4n$$

$$m - 3n < n, \quad 11n - 3m < m$$

$$\text{let } m - 3n = p, \quad 11n - 3m = q, \quad p, q \in \mathbb{N}$$

$$\frac{q}{p} = \sqrt{11}, \quad \text{this process can be infinitely repeated}$$

$$\text{concludes with } m, n, p, q \in \mathbb{N}$$

$$\text{so } \sqrt{11} \text{ irrational}$$

$$\text{for } \sqrt{11}, \quad 10 < \sqrt{11} < 11$$

$$\text{if } \frac{m}{n} = \sqrt{11}, \quad \frac{11n - 10m}{m - 10n} = \sqrt{11}$$

$$\text{for } \sqrt{21}, \quad 10 < \sqrt{21} < 12$$

we cannot deduce the new denominator is smaller than the original denominator

therefore method fails

0). $r + \alpha$ irrational

α cannot be written as $\frac{p}{q}$, $p, q \in \mathbb{N}$

so $r + \alpha$ cannot be written as $\frac{m}{n}$, $m, n \in \mathbb{N}$

hence irrational

- $r\alpha$ can be rational, let $r=0$
- $\alpha + \beta$ can be rational, let $\alpha = -\beta$
- $\alpha\beta$ can be rational, let $\alpha = \beta = \sqrt{2}$
- r^α can be rational, let $r=3$, $\alpha = \log_3 4$
- α^r can be rational, let $r=0$
- α^β can be rational, let $\alpha=e$, $\beta=\ln 3$

8)

i) consider pairs $(1, 2), (3, 4), \dots, (2n-1, 2n)$

2 consecutive integers are coprime

statement proved by pigeonhole principle

ii) consider pairs $(1, 2n), (2, 2n-1), \dots, (n, n+1)$

then use pigeonhole principle

iii) consider different sets of numbers in the form $2^n, 3 \cdot 2^n, 5 \cdot 2^n, 7 \cdot 2^n, \dots$

n sets of these, then use pigeonhole principle

iv) for a set with $n+1$ elements, it produces n distinct difference,

then use pigeon hole principle, first n elements can't all be distinct

but the $n+1$ th element must be distinct

counter examples:

i) n even number

ii) n odd number

iii) $\{1, 2, 3, 4\}$, $S = \{2, 3\}$

iv) n odd number

no such example where all 4 fails

9)

A: B ~ F 5

B: C ~ F 4

C: D ~ F 3

D: E ~ F 2

E: F 1

15 edges

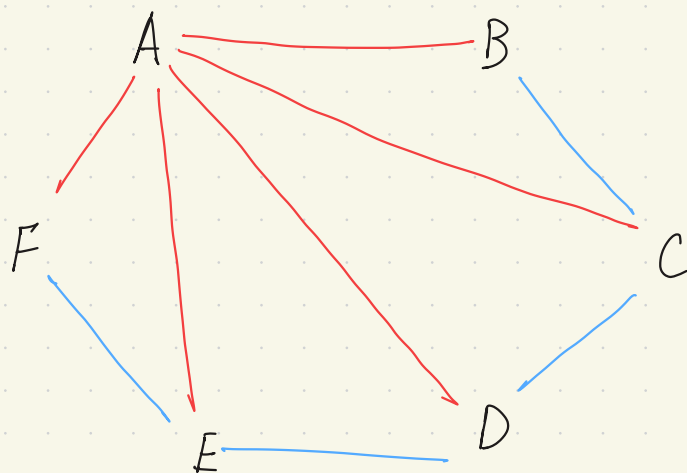
at least 8 edges are the same transport

try to place the 8 edges without forming a triangle

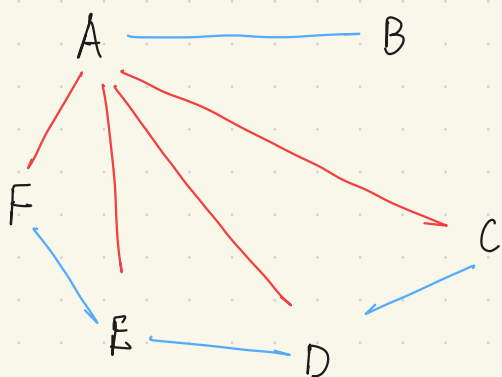
Assume AB, CD, EF are edges that don't meet each other

AC, AD, AE, AF

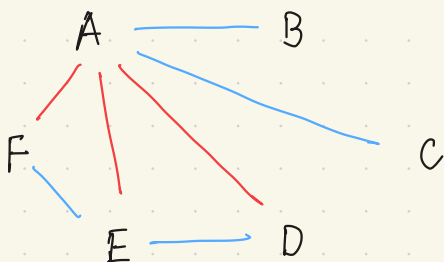
IGNORE ABOVE



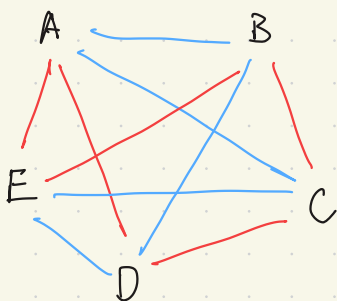
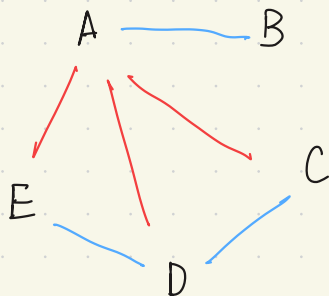
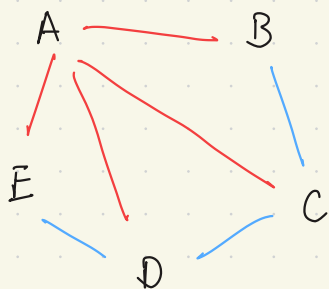
FD, EC, BD
can be any colour



FD, EC can be any colour



FD can be any colour



not true if only 5 towns